Deliverable 3
Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Introduction

Time-series forcasting relevance and a brief description of the traditional methods; the deep-learning tecniques including fine-tuned architecture and transformers based ones. A brief description of the Chronos family.A simple introduction on signal pre-processing, the Nyquist-Shannon sampling theorem and the aliasing phenomenon.

Patch-Based Tokenization: a Formal Description

Formalization of Patch-Based Tokenization

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the *patch-based tokenization operator* $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is formally defined as:

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix} \tag{1}$$

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.1 Pro, Gemini 3.5 Flash, Sonnet 4.6, and Opus 4.7, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Using an element-wise indexing notation where $\mathbf{t}_k[m]$ denotes the m-th component of the k-th token (for $m \in \{0, 1, \ldots, P-1\}$):

$$\mathbf{t}_k[m] = x[kS + m] \tag{2}$$

Derivation of Patch Indistinguishability Conditions

We now derive the conditions under which two consecutive tokens, $\mathbf{t}_k$ and $\mathbf{t}_{k+1}$, become indistinguishable to the model architecture. By definition, consecutive tokens are identical if and only if $\mathbf{t}_k[m] = \mathbf{t}_{k+1}[m]$ for all $m \in \{0, 1, \ldots, P-1\}$. Substituting Equation (2), this yields the system of equations:

$$x[kS + m] = x[(k+1)S + m], \quad \forall m \in \{0, 1, \ldots, P-1\} \tag{3}$$

Letting $n = kS + m$, Equation (3) states that the signal must satisfy:

$$x[n] = x[n + S] \tag{4}$$

for all time indices n spanned by the consecutive patches, namely $n \in \{kS, kS+1, \ldots, kS+P+S-1\}$.

If the underlying discrete signal $\mathbf{x}$ is periodic or contains a dominant periodic component with fundamental period N_0, then Equation (4) is satisfied if the tokenization stride S is an integer multiple of the signal's period:

$$S = c \cdot N_0, \quad c \in \mathbb{N}^+ \tag{5}$$

In terms of normalized angular frequency $\omega_0 = \frac{2\pi}{N_0}$, this condition translates to:

$$\omega_0 = \frac{2\pi c}{S} \tag{6}$$

When Equation (6) holds, the stride perfectly matches the signal's structural frequency, causing every consecutive patch to capture the exact same phase profile of the wave, rendering $\mathbf{t}_k = \mathbf{t}_{k+1}$.

Characterization of the Loss of Observability

When consecutive patches become indistinguishable ($\mathbf{t}_k = \mathbf{t}_{k+1}$), a severe *structural aliasing* event occurs, leading to a catastrophic loss of observability in the token space. We characterize this degradation through three lenses:

1. **Rank Deficiency in the Embedding Space:** Let $\mathbf{M} = [\mathbf{t}_k, \mathbf{t}_{k+1}, \ldots, \mathbf{t}_{k+L-1}] \in \mathbb{R}^{P \times L}$ be the local trajectory matrix constructed by the tokenization layer. If Equation (4) holds, then $\mathbf{t}_k = \mathbf{t}_{k+1} = \cdots = \mathbf{t}_{k+L-1}$. The column rank of $\mathbf{M}$ collapses to 1:

$$\text{rank}(\mathbf{M}) = 1 \tag{7}$$

   Consequently, the token sequence loses all geometric variance across time steps.
2. **Attention Matrix Degeneration:** Modern foundation models rely on self-attention mechanisms. The raw attention score between two token representations $\mathbf{h}_k$ and $\mathbf{h}_j$ (derived from $\mathbf{t}_k$ and $\mathbf{t}_j$) is computed via a dot-product. If $\mathbf{t}_k = \mathbf{t}_{k+1}$, their linear projections are identical ($\mathbf{q}_k = \mathbf{q}_{k+1}$ and $\mathbf{k}_k = \mathbf{k}_{k+1}$). The attention weight matrix $\mathbf{A}$ degenerates into uniform rows:

$$A_{k,j} = A_{k+1,j}, \quad \forall j \tag{8}$$

   The model's causal mask and positional encodings are forced to carry the entire burden of temporal differentiation, completely isolating the model from the underlying signal dynamics.
3. **Phase and Higher-Frequency Masking:** Because the operator $\mathcal{T}_{P,S}$ samples the state space at macro-intervals of S, any high-frequency oscillations or intra-patch phase variations occurring at frequencies $\omega > \frac{\pi}{S}$ are completely folded back (aliased) into a static DC component ($\omega_{\text{alias}} = 0$). The model perceives a completely stationary system, rendering the true underlying temporal evolution unobservable.

Patch-Based Tokenization: Claude

Setup and Formal Definition of the Tokenisation Operator

To rigorously understand the failure modes of patch-based architectures like Chronos-Bolt, we must first formally define the tokenisation process. Let $x[n]$ represent a discrete-time, univariate signal where the index $n \in \mathbb{Z}$ is sampled at a continuous frequency of f_s.

Unlike classical autoregressive models that process temporal data one scalar time-step at a time, modern foundation models employ patch-based tokenisation. This technique groups contiguous samples to radically reduce the effective sequence length and improve the computational efficiency of the transformer's attention mechanisms. This is achieved by sliding a window of fixed size P (the patch length) across the signal, moving forward by a step size of S (the stride).

We can extract this sequence of tokens as follows:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \ldots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \quad k \in \mathbb{N} \tag{9}$$

Definition 1 (Patch Tokenisation Operator)**.** *We can formalise this extraction as a linear map acting on the bounded signal space. The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$, as defined above, is referred to as the* patch tokenisation operator*. When collecting an arbitrary number of K patches from a signal vector $\mathbf{x}$, the operation can be represented as a matrix multiplication:*

$$\mathbf{P} = \mathcal{T}\{x\} = \mathbf{W} \cdot \mathbf{x} \tag{10}$$

Here, $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a block-binary selection matrix. Its k-th block consists of P rows that act as a mask, picking out the exactly P consecutive signal samples starting at the temporal index kS.

The Mechanics of Phase-Locking and Degeneracy

The fundamental vulnerability of the patch tokenisation operator emerges when the periodic structure of the input signal aligns perfectly with the model's fixed stride grid. We refer to this specific synchronization as "phase-locking."

Proposition 1 (Condition for Token Identity)**.** *Two consecutive patches, $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$, will be mathematically identical if and only if the signal amplitude remains constant across the stride shift for every element within the observation window:*

$$x[n] = x[n+S] \quad \forall n \in \{kS,\, kS+1,\, \ldots,\, kS+P-1\} \tag{11}$$

Globally, if the underlying continuous signal is a pure sinusoid (or any periodic signal) with a fundamental period of T_0 samples, this exact alignment occurs whenever the stride S spans an integer multiple of the signal's period:

$$\boxed{S = m \cdot T_0, \quad m \in \mathbb{Z}^+} \tag{12}$$

By substituting $T_0 = f_s/f$, we can express this condition in terms of the signal's continuous frequency f and the sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \quad m \in \mathbb{Z}^+ \tag{13}$$

Proof. To satisfy $\mathbf{p}_k = \mathbf{p}_{k+1}$ strictly component-wise, it is required that $x[kS+j] = x[(k+1)S+j]$ for every internal patch index $j \in \{0, \ldots, P-1\}$. For a periodic signal defined by $T_0 = f_s/f$, this strict equality holds true whenever the stride shift S represents a traversal of exactly m full wave cycles. In such instances, the sliding window captures the exact same geometrical segment of the wave at every step. □

The Consequence: Loss of Observability

When the phase-locking condition $f = m \cdot (f_s/S)$ is met, we achieve a state where $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all possible values of k. The token matrix $\mathbf{P}$ undergoes a mathematical rank collapse, reducing its rank to exactly 1.

This creates a "degenerate token sequence", which induces severe representational failures within the network:

- **Complete Temporal Blindness:** Because the transformer processes a sequence of completely identical vectors, the temporal variation of the actual underlying signal becomes entirely invisible to the model's self-attention mechanisms. The model cannot perceive that a wave is oscillating; it simply perceives a constant, flat embedding.
- **Observational Equivalence and Aliasing:** Any two distinct signals that differ only by a phase shift of S samples will produce the exact same sequence of tokens. They become observationally indistinguishable to the decoder, making it impossible for the model to correctly reconstruct or forecast the true phase of the original signal.
- **Soft Degradation at Boundary Frequencies:** This failure is not isolated to exact integer multiples. Even when patches are not perfectly identical but are merely highly correlated (i.e., when $S \approx mT_0$), the sequence suffers a *soft* loss of observability. The effective information bandwidth is severely restricted, explaining why predictive accuracy degrades smoothly around these critical frequency boundaries before fully collapsing.

For example, in the specific architecture of Chronos-Bolt, the sampling frequency is $f_s = 512$ Hz and the stride is $S = 16$. According to our derived condition $f = m \cdot (512/16)$, the fundamental patch frequency is precisely 32 Hz. Consequently, any input signal at 32 Hz, 64 Hz, 96 Hz, or other integer harmonics will trigger this precise structural aliasing, resulting in an immediate and sharp collapse in the model's predictive accuracy.

Phase-Locking and Token Space Representation

Generalized Mathematical Framework of Patch-Stride Phase-Locking

Consider a continuous-time periodic signal sampled uniformly at a sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate and model a fundamental harmonic component of the signal as a discrete sinusoid:

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) \tag{14}$$

where f denotes the signal frequency satisfying the strict Nyquist criterion ($f < \frac{f_s}{2}$), and $\phi \in [0, 2\pi)$ represents an arbitrary initial phase offset.

Derivation of the Phase-Locking Frequency Class Phase-locking within a grid-based windowing architecture occurs when the structural phase profile of the signal repeats exactly across uniform translational shifts equal to the operational stride S.

Evaluating a forward translation of the sequence by a distance equal to the stride S yields:

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{f \cdot S}{f_s} + \phi\right) \tag{15}$$

For the signal to exhibit absolute periodic alignment with the tokenization grid, the phase accumulation over the interval S must equal an integer multiple of a full rotation (2π). Thus, we establish the boundary condition:

$$2\pi \frac{f \cdot S}{f_s} = 2\pi k, \quad k \in \mathbb{N}^+ \tag{16}$$

Solving Equation (3) explicitly for the signal frequency f characterizes the exact class of phase-locking frequencies, denoted as $\mathcal{F}_{\text{lock}}$:

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = k \cdot \frac{f_s}{S},\ k \in \mathbb{N}^+ \right\} \tag{17}$$

Defining the fundamental grid frequency as $f_{\text{grid}} = \frac{f_s}{S}$, the phase-locking condition simplifies to $f = k \cdot f_{\text{grid}}$. This proves that phase-locking is an architectural consequence governed entirely by the harmonic alignment between the signal frequency and the stride-based grid frequency.

Analysis of Structural Degeneracies and Invariances

When the input signal frequency belongs to the phase-locking class ($f \in \mathcal{F}_{\text{lock}}$), the resulting impact on the vector-valued token sequence $\mathbf{T} = \{\mathbf{t}_j\}_{j\in\mathbb{Z}}$ (where $\mathbf{t}_j \in \mathbb{R}^P$) is determined by the geometric interplay between P and S.

Case 1: Equal Patch Size and Stride ($P = S$) When the windowing process is strictly contiguous and non-overlapping ($P = S$), the phase-locking condition matches the patch size ($f = k\frac{f_s}{P}$). Substituting this condition into the component-wise definition of the token vector $\mathbf{t}_j[m] = x[jP + m]$ for $m \in \{0, 1, \ldots, P-1\}$ yields:

$$\mathbf{t}_{j+1}[m] = x[(j+1)P + m] = x[jP + m + P] \tag{18}$$

Applying the grid invariance condition $x[n + P] = x[n]$ derived from Equation (4):

$$\mathbf{t}_{j+1}[m] = x[jP + m] = \mathbf{t}_j[m] \tag{19}$$

Consequently, $\mathbf{t}_{j+1} = \mathbf{t}_j$ for all sequential steps j. This induces a **complete token-level degeneracy**. The sequence of representations collapses into a static series of uniform vectors, eradicating all observable temporal velocity and dynamic gradients for downstream sequence models.

Case 2: Overlapping Boundaries ($S < P$) In an overlapping configuration, the stride is smaller than the patch length. However, because the phase-locking condition $\mathcal{F}_{\text{lock}}$ is strictly dependent on the stride S, the identity $x[n + S] = x[n]$ remains globally valid across all samples. Evaluating the sequential tokens:

$$\mathbf{t}_{j+1}[m] = x[(j+1)S + m] = x[jS + m + S] = x[jS + m] = \mathbf{t}_j[m] \tag{20}$$

Even though individual patches span a wider historical context (P), the underlying signal periodicity matches the stride interval perfectly. Thus, consecutive tokens remain structurally identical ($\mathbf{t}_{j+1} = \mathbf{t}_j$), demonstrating that decreasing the stride frequency below the signal frequency preserves token space degeneracy regardless of overlap size.

Case 3: Disjoint / Underlapping Windows ($S > P$) When $S > P$, spatial gaps occur between successive windows. Under the condition $f \in \mathcal{F}_{\text{lock}}$, the relation $x[n + S] = x[n]$ still ensures that $\mathbf{t}_{j+1} = \mathbf{t}_j$. However, this scenario introduces an additional information risk: the unobserved intervals of length $S - P$ are omitted entirely from tokenization. High-frequency phase variations occurring within those hidden intervals are completely masked, leading to severe under-sampling and a permanent loss of structural observability.

Domain Description and Ontology Representation

Simple domain description and reference to the ontology representation using .ttl as an appendix.

Structural Aliasing Expected Impact

Formalization of Falsifiable Hypotheses

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate f_s (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size P (`pv:patchSize`) and translation stride S (`pv:stride`).

The architectural grid frequency f_p (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \tag{21}$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

Hypothesis 1 (H_1): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules `R18` and `R19`).
- **Statement:** When the fundamental frequency f (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency f_p, the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

  $$f = n \cdot f_p = n\left(\frac{f_s}{S}\right), \quad n \in \mathbb{N}^+ \tag{22}$$

  Under this boundary condition, the column rank of the local trajectory matrix collapses to unity, rendering the internal dynamics of the patch unobservable to downstream self-attention mechanisms.
- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band $[2, \frac{f_s}{2}]$ Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of f_p.

Hypothesis 2 (H_2): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule `R20`).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal's peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency $f \in \mathcal{F}_{\text{lock}}$, translating the initial phase offset ϕ (`pv:phaseRad`) within the discrete-time sinusoid:

  $$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \quad \phi \in [0, 2\pi) \tag{23}$$

  modifies the specific sub-sequence frozen inside each individual patch window. If the phase shift satisfies $\phi \in (0, S)$, it induces a structural misalignment that alters the component-wise values of consecutive token vectors, systematically modulating the token space fidelity.
- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric SV) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements $\phi \sim \text{Uniform}(0, 2\pi)$ at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

Hypothesis 3 (H_3): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule `R21`).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.
- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio or (`pv:overlapRatio`):

$$or = \frac{P - S}{P} \tag{24}$$

  When $or = 0$ ($P = S$), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ($\mathbf{t}_j = \mathbf{t}_{j+1}$), wiping out temporal gradients. When $or \to 1$ ($S \ll P$), successive patches share massive historical context; although the stride-bound phase-locking condition $x[n + S] = x[n]$ still holds, the expanded patch dimension P forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.
- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters P and S while keeping a locked signal frequency f yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

Signal Generation

Simple sinusoid generation. TSMixup and KernelSynth description, algorithm pseudo-code.